

THE HONG KONG POLYTECHNIC UNIVERSITY
DEPARTMENT OF APPLIED MATHEMATICS
AMA3640 STATISTICAL INFERENCE

Assignment 3

Due at 11:59pm on 4 Dec, 2025

1. Let X be a random variable with distribution

$$P_\theta(X \leq x) = 1 - e^{-(x/\theta)^2} \quad \text{for } x \geq 0,$$

where $\theta > 0$.

- (a) Consider testing $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$ for some positive constant θ_0 . We reject H_0 if $X > k$. What choice of k would result in a size α test?
 - (b) Consider the class of tests given in (a). Suppose that $\theta_0 = 2$ and the observed value of X is 0.971. What is the observed p -value?
 - (c) Construct an 80% confidence interval for θ that takes the form of $(0.2X/c, X/c)$ for some positive constant c . What value can c take? You can use the fact that $(0.8043)(1 - 0.8043^{24}) = 0.8$.
2. Let $X_1, \dots, X_n$ be a random sample from $\text{Gamma}(2, \theta)$, where the density is

$$f(x; \theta) = \theta^2 x e^{-\theta x} \quad \text{for } x > 0,$$

where $\theta > 0$. Let $Y_1, \dots, Y_m$ be a random sample from the exponential distribution with density $f(y; \lambda) = \lambda e^{-\lambda y}$ for $y > 0$, where $\lambda > 0$; X_i 's and Y_j 's are independent.

- (a) Construct a size α test for $H_0 : \theta/\lambda = \tau_0$ versus $H_1 : \theta/\lambda \neq \tau_0$ for some positive constant τ_0 . Express the rejection region in terms of quantiles of an F distribution. You can use the fact that $\sum_{i=1}^n X_i \sim \text{Gamma}(2n, \theta)$.
 - (b) Derive a $(1 - \alpha)$ confidence interval for θ/λ .
3. Let $X_1, \dots, X_n$ be a random sample from the exponential distribution with rate $1/\theta$, where the density is

$$f(x | \theta) = \frac{1}{\theta} e^{-x/\theta} \quad \text{for } x > 0.$$

Let $\hat{\theta}_1 = \bar{X}$ and $\hat{\theta}_2 = n\bar{X}/(n+1)$. Consider the squared error loss.

- (a) Evaluate the risk functions of $\hat{\theta}_1$ and $\hat{\theta}_2$ (as estimators of θ).
 - (b) Set the prior distribution of θ to be exponential with rate 0.3. That is, $\pi(\theta) = 0.3e^{-0.3\theta}$ for $\theta > 0$. Find the Bayes risk of $\hat{\theta}_1$.
4. Let $X_1, \dots, X_n$ be a random sample from the $\text{Gamma}(3, \theta)$ distribution with $\theta > 0$, where the density is

$$f(x | \theta) = \frac{\theta^3}{2} x^2 e^{-\theta x} \quad \text{for } x > 0.$$

The prior distribution of θ is exponential with rate β , that is, $\pi(\theta) = \beta e^{-\beta\theta}$ for $\theta > 0$.

- (a) Derive a $(1 - \alpha)$ credible interval for θ . Give the interval in terms of quantiles of a chi-square distribution.
 - (b) Derive a $(1 - \alpha)$ credible interval for $1/\theta$.
5. A random sample, $X_1, \dots, X_5$, has been taken from a $\text{Uniform}(0, \theta)$ distribution, with density

$$f(x | \theta) = \frac{1}{\theta} \quad \text{for } 0 < x < \theta.$$

Adopt a Bayesian paradigm, and set the density of the prior distribution of θ to be

$$\pi(\theta) = \frac{\alpha\beta^\alpha}{\theta^{\alpha+1}} \quad \text{for } \theta > \beta,$$

with $\alpha = 1$ and $\beta = 2$. The observed values of $X_1, \dots, X_5$ are

0.52, 1.25, 1.78, 2.15, 2.40.

- (a) Derive the density of the posterior distribution of θ given the observed data. (Hint: $\pi(\theta)$ given above is a density for any $\alpha > 0$ and $\beta > 0$.)
- (b) Find the posterior expected value of θ .
- (c) Construct a 95% credible interval for θ .

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Assignment 3 Solution

1. (a) The power function is $\beta(\theta) = P(X > k) = e^{-(k/\theta)^2}$, which is increasing in θ . Therefore, $\sup_{\theta \leq \theta_0} \beta(\theta) = \beta(\theta_0)$, and the size of the test is $e^{-(k/\theta_0)^2}$. To obtain a size α test, we set $e^{-(k/\theta_0)^2} = \alpha$, which yields $k = \theta_0(-\log \alpha)^{1/2}$.
- (b) The p -value is

$$P(X > 0.971; \theta = 2) = e^{-(0.971/2)^2} = 0.79.$$

- (c) Note that

$$\begin{aligned} P\left(\frac{0.2X}{c} < \theta < \frac{X}{c}\right) &= P(c\theta < X < 5c\theta) \\ &= e^{-(c\theta/\theta)^2} - e^{-(5c\theta/\theta)^2} \\ &= e^{-c^2} - e^{-25c^2} \\ &= e^{-c^2} \{1 - (e^{-c^2})^{24}\}. \end{aligned}$$

We wish to find c such that the above is equal to 0.8. One solution is that $e^{-c^2} = 0.8043 \implies c = 0.4667$. Therefore, an 80% confidence interval for θ is $(0.4286X, 2.1428X)$.

2. (a) Note that

$$\sum_{i=1}^n X_i \sim \text{Gamma}(2n, \theta) \iff 2\theta \sum_{i=1}^n X_i \sim \text{Gamma}(2n, 1/2) = \chi_{4n}^2.$$

Also, $2\lambda \sum_{j=1}^m Y_j \sim \text{Gamma}(m, 1/2) = \chi_{2m}^2$. Therefore,

$$\frac{2\theta \sum_{i=1}^n X_i / (4n)}{2\lambda \sum_{j=1}^m Y_j / (2m)} = \frac{m\theta \sum_{i=1}^n X_i}{2n\lambda \sum_{j=1}^m Y_j} \sim F_{4n, 2m}.$$

Under $H_0 : \theta/\lambda = \tau_0$,

$$P\left(F_{4n, 2m, 1-\alpha/2} < \frac{m\tau_0 \sum_{i=1}^n X_i}{2n \sum_{j=1}^m Y_j} < F_{4n, 2m, \alpha/2}\right) = 1 - \alpha.$$

A possible size α test is to reject H_0 if and only if

$$\frac{m\tau_0 \sum_{i=1}^n X_i}{2n \sum_{j=1}^m Y_j} < F_{4n, 2m, 1-\alpha/2} \quad \text{or} \quad \frac{m\tau_0 \sum_{i=1}^n X_i}{2n \sum_{j=1}^m Y_j} > F_{4n, 2m, \alpha/2}.$$

- (b) A $(1 - \alpha)$ confidence interval for θ/λ can be obtained by inverting the test in (a):

$$P_\tau \left(F_{4n,2m,1-\alpha/2} < \frac{m\tau \sum_{i=1}^n X_i}{2n \sum_{j=1}^m Y_j} < F_{4n,2m,\alpha/2} \right) = 1 - \alpha$$

$$\iff P_\tau \left(\frac{2n}{m} F_{4n,2m,1-\alpha/2} \frac{\sum_{j=1}^m Y_j}{\sum_{i=1}^n X_i} < \tau < \frac{2n}{m} F_{4n,2m,\alpha/2} \frac{\sum_{j=1}^m Y_j}{\sum_{i=1}^n X_i} \right) = 1 - \alpha.$$

Therefore, a $(1 - \alpha)$ confidence interval for θ/λ is

$$\left(\frac{2n F_{4n,2m,1-\alpha/2}}{m} \frac{\sum_{j=1}^m Y_j}{\sum_{i=1}^n X_i}, \frac{2n F_{4n,2m,\alpha/2}}{m} \frac{\sum_{j=1}^m Y_j}{\sum_{i=1}^n X_i} \right).$$

3. (a) Clearly, $\hat{\theta}_1$ is unbiased, so its risk function is

$$\text{Var}(\hat{\theta}_1) = \frac{\theta^2}{n}.$$

The risk function of $\hat{\theta}_2$ is

$$\text{Var}(\hat{\theta}_2) + \text{bias}(\hat{\theta}_2)^2 = \frac{n\theta^2}{(n+1)^2} + \frac{\theta^2}{(n+1)^2} = \frac{\theta^2}{n+1}.$$

- (b) The prior expectation of θ^2 is

$$\int_0^\infty 0.3\theta^2 e^{-0.3\theta} d\theta = \frac{2}{0.3^2} = \frac{200}{9}.$$

Therefore, the Bayes risk of $\hat{\theta}_1$ is

$$E \left(\frac{\theta^2}{n} \right) = \frac{200}{9n}.$$

4. (a) The posterior density of θ is

$$\pi(\theta | \mathbf{x}) \propto e^{-\beta\theta} \left(\prod_{i=1}^n \theta^3 e^{-\theta x_i} \right) = \theta^{3n} e^{-\theta(\beta + \sum_{i=1}^n x_i)},$$

so that $\theta | \mathbf{X} \sim \text{Gamma}(3n + 1, \sum_{i=1}^n X_i + \beta)$. The posterior distribution of $2(\sum_i X_i + \beta)\theta$ is $\chi_{2(3n+1)}^2$. A $(1 - \alpha)$ credible interval for θ is

$$\left(\frac{\chi_{2(3n+1),1-\alpha/2}^2}{2(\sum_{i=1}^n X_i + \beta)}, \frac{\chi_{2(3n+1),\alpha/2}^2}{2(\sum_{i=1}^n X_i + \beta)} \right).$$

- (b) A $(1 - \alpha)$ credible interval for $1/\theta$ is

$$\left(\frac{2(\sum_{i=1}^n X_i + \beta)}{\chi_{2(3n+1),\alpha/2}^2}, \frac{2(\sum_{i=1}^n X_i + \beta)}{\chi_{2(3n+1),1-\alpha/2}^2} \right).$$

5. (a) The posterior density is proportional to the likelihood times the prior:

$$\pi(\theta|\mathbf{x}) \propto f(\mathbf{x}|\theta)\pi(\theta) = \frac{1}{\theta^n} I(\max x_i < \theta) \cdot \frac{\alpha\beta^\alpha}{\theta^{\alpha+1}} I(\theta > \beta) \propto \frac{1}{\theta^{n+\alpha+1}} I(\theta > \max\{\beta, \max x_i\}).$$

With $\alpha = 1$, $\beta = 2$, $n = 5$, and $\max x_i = 2.40$, we have

$$\pi(\theta|\mathbf{x}) \propto \frac{1}{\theta^7} I(\theta > 2.4).$$

Since the posterior density integrates to one, we have

$$\pi(\theta|\mathbf{x}) = \frac{1146.618}{\theta^7}, \quad \theta > 2.4.$$

- (b) The posterior mean is

$$E(\theta|\mathbf{x}) = \int_{2.4}^{\infty} \theta \cdot \frac{1146.618}{\theta^7} d\theta = 1146.618 \int_{2.4}^{\infty} \frac{1}{\theta^6} d\theta = 2.88.$$

- (c) To find a 95% credible interval for θ , solve for l and u such that

$$\int_l^{\infty} \pi(\theta|\mathbf{x}) d\theta = 0.975, \quad \int_u^{\infty} \pi(\theta|\mathbf{x}) d\theta = 0.025.$$

This gives $l = 2.410$ and $u = 4.438$. Therefore, a 95% credible interval for θ is $(2.410, 4.438)$.